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Experimental study of fractional-order models for lithium-ion battery and ultra-capacitor: Modeling, system identification, and validation☆

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HIGHLIGHTS

- Experiments of electrochemical impedance spectroscopy are performed.
- Effects of temperature and SOC on impedance spectra are analyzed.
- Fractional-order models for lithium-ion batteries and ultra-capacitors are built.
- The global optimization is proposed for model parameter identification.
- The model accuracy is proved by experiments at different temperatures.

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ABSTRACT

In recent years, with the promotion of new energy vehicles and micro-grids, energy storage technology has achieved considerable development. The hybridization of the lithium-ion batteries with ultra-capacitors can effectively improve the power capability of the energy storage system, which is suitable for the vehicle and micro-grid applications. High precision modeling is the foundation of the state estimation and energy management of the energy storage system. The fractional-order model provides applicable computational complexity and accuracy. This paper presents the experimental study of the fractional-order models used for lithium-ion batteries and ultra-capacitors. In this work, the experimental results of the electrochemical impedance spectroscopy of both lithium-ion batteries and ultra-capacitors are systematically analyzed. Then the temperature compensating fractional-order models of the batteries and ultra-capacitors are developed. Moreover, the particle swarm optimization based global optimization is proposed for on-line parameter identification with strict constraints. Therefore the parameters of the fractional-order models can be extracted from daily vehicle operating conditions. Finally, the accuracy of the presented model is proved by dynamic cycle conditions at different temperatures. The results indicate that the proposed method can well approximate the voltage of both the lithium-ion batteries and the ultra-capacitors with mean relative errors less than 4% and 3%, respectively.

1. Introduction

Efficient energy storage plays a core role in the development of the strategic emerging industries of energy conservation and new energy technology. In recent years, the deficiency of power density and cycle life of lithium-ion batteries directly restrict the development of energy storage technology. Compared with the lithium-ion batteries, the ultra-capacitors have relatively high power density and long cycle life, which are suitable for the rapid and frequent charge/discharge [1]. The hybridization of lithium-ion batteries with ultra-capacitors can effectively solve the drawbacks of the lithium-ion batteries, and maintain the energy output of the system [2]. Accurate modeling is the basis of state estimation and power management of large-scale energy storage systems [3]. Therefore, this paper systematically presents the methodology of frequency-domain and time-domain modeling, system identification, and validation of the fractional-order models used for lithium-ion batteries and ultra-capacitors. The purpose of this paper is to provide some inspirations for the design, control, and management of hybrid energy storage systems.

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Nomenclature

Acronyms & Abbreviations

PDE	Partial differential equation
ECM	Equivalent circuit model
EM	Electrochemical model
EIS	Electrochemical impedance spectroscopy
CPE	Constant phase element
EDLC	Electronic double-layer capacitors
AC	Alternating current
SOC	State-of-charge
OCV	Open-circuit voltage
PSO	Particle swarm optimization
MAE	Mean absolute error
MRE	Mean relative error
RMSE	Root-mean-square error

Notation

Z	Impedance
V	Terminal voltage
I	Current
T	Temperature
z	State-of-charge
f	Frequency
V_{oc}	Open-circuit voltage
k_i (i=0, 1, …, 5)	Polynomial parameters of OCV-SOC function
R_0	Ohmic internal resistance of the lithium-ion battery
R_1	Polarization internal resistance of the lithium-ion battery
Z_{CPE}	Impedance of CPE
$Z_{Warburg}$	Impedance of Warburg-like component
R_s	Ohmic internal resistance of the ultra-capacitor
R_2	Polarization internal resistance of the ultra-capacitor
C	Capacitive reactance property of CPE
A	Fractional order of CPE
W	Capacitive reactance property of Warburg-like component
B	Fractional order of Warburg-like component
$\mathfrak{D}^{\mu}$	Operator of fractional-order calculus
μ	Integral-differential order
$\triangle t$	Sampling time
I_1	Current of the battery
I_2	Current of the ultra-capacitor
Y_1	Difference between battery OCV and terminal voltage
Y_2	Terminal voltage of the ultra-capacitor
N	Length of the training dataset
$Nfom$	Length of fractional-order calculus
$\boldsymbol{\theta}$	Parameter vectors

For the system modeling of lithium-ion batteries and ultra-capacitors, plenty of modeling methods have been proposed. In Ref. [4], the Thevenin model and a standard RC model are established for battery and ultra-capacitor, and a co-estimator for parameter and state joint estimation is proposed. Tang et al. [5] proposed a new migrated model for high power Li-ion batteries considering the uncertainties of temperature and aging. Ref. [6] presented a full cell model including the lithium anode, solid polymer electrolyte, and insertion composite cathode. Ref. [7] developed a framework for simplification of partial differential equation (PDE) based lithium-ion battery models. The obtained libraries of simplified models can be used in various applications towards advanced battery management. Generally those models can be divided into different categories according to the requirements of model complexity and accuracy in different application scenarios [8]. The equivalent circuit models (ECMs) are the most popular models used in engineering applications because they are simple and can approximate the input/output behaviors well [9]. The electrochemical models (EMs) have enough simulation accuracy and can predict internal cell electrochemical variables. The evolution of these electrochemical variables provides strong inspiration and reference for premature aging prediction and health prognostic of the energy storage devices [10]. However, these models should be simplified in order to be better applied for online parameter identification and state estimation in real application [11].

It is well known that an ECM with well-chosen component values can closely approximate the behaviors of an energy storage device. For the conventional ECMs, the resistor–capacitor (RC) networks are usually used to approximate the dynamic processes such as polarization and diffusion phenomena. However, if electrochemical impedance spectroscopy (EIS) is used to analyze the characteristic of the energy storage device, this kind of modeling will be proved defective [12]. For example, in the middle-frequency, the battery's Nyquist curve is not a standard semicircle, which means that the standard RC network is not appropriate in this case. One approach for solving this problem is to use the constant phase elements (CPEs) instead of the capacitors in RC networks [13]. This comes out the fractional-order models. Ref. [14] provided an overview of the current developments in circuit-based models for batteries and ultra-capacitors. The links among the fractional-order calculus, electrochemical impedance spectroscopy and dynamic characteristics of electrochemical energy systems are illustrated in detail. Yang et al. [15] presented a simplified fractional order impedance model for the lithium-ion batteries, and the least square genetic algorithm was used to identify the model parameters. The maximum voltage error for the proposed model was within 0.5%. Gabano et al. [16] reported an impedance identification approach for ultra-capacitors models by using linear parameter varying fractional modeling which can properly estimate non-linear high-level voltage dynamics. Kumar et al. [17] presented the frequency-dependent piecewise fractional-order modeling method for ultra-capacitors. To maintain the trade-off between model complexity and accuracy, the chaotic seeker optimization algorithm was used for parameter identification.

At present, the fractional-order methodology has been used not only for high accuracy modeling of the energy storage systems [18,19], but also for state estimation [20], health prognosis [21], and fault diagnosis [22]. In Ref. [23], the remaining discharge time of battery and ultra-capacitor are predicted based on fractional-order models and Bayesian method. Zhang et al. [24] presented a fractional-order model for ultra-capacitors and a fractional Kalman filter was designed to estimate the SOC of the ultra-capacitor. Tian et al. [25] constructed a framework of battery degradation state recognition based on fractional-order model and incremental capacity analysis. Ref. [26] indicated that there was a deterministic mapping relationship between the fractional order and the battery cycle numbers. Moreover, when the fractional order tended to be stable, the electrode life was over. Zhang et al. [27] compared the precision between a fractional-order model and a Thevenin model at different temperature and SOC. Then a battery external short circuit fault diagnosis approach was proposed based on random forests algorithm and the fractional-order model. However, in order to further improve the fidelity of the fractional-order model, the impact of temperature should be considered in the process of modeling. Besides, there are few researches on the application of fractional-order model in hybrid energy storage system.

In this work, the fractional-order modeling of the lithium-ion batteries and ultra-capacitors, and the parameter identification approaches in frequency and time domain are presented. First, the EIS testing of the lithium-ion batteries and ultra-capacitors are performed and the Nyquist

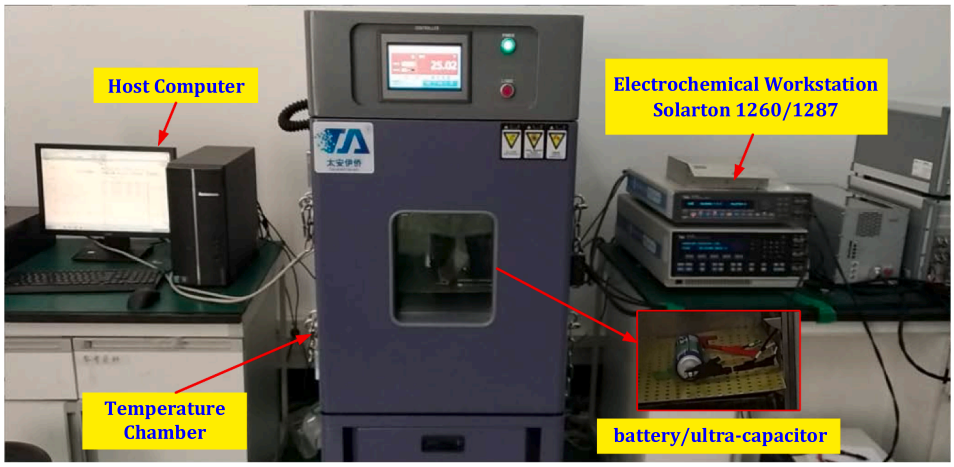

Fig. 1. Experimental setup for impedance frequency response data acquisition.

curves in the frequency domain are plotted in order to determine the model structure of the energy storage device. Second, the temperature compensating fractional-order models for the lithium-ion batteries and ultra-capacitors are developed. Then the parameter fitting results in the frequency domain are presented. To better apply the presented models in a real system, the parameter identification in time domain using global optimization is introduced. Third, the model estimation results at different temperatures are compared and analyzed. The results of time domain identification are compared with the results of frequency domain fitting. It indicates that the proposed temperature compensating fractional-order models can well approximate the voltage of both the lithium-ion batteries and the ultra-capacitors.

The remainder of the paper is organized as follows: Section 2 describes the experimental study of EIS for the batteries and ultra-capacitors. Section 3 provides details about the mathematical deduction of the fractional-order models for batteries and ultra-capacitors. Moreover, the parameter identification in time domain using global optimization is introduced. Experiments and results of the time domain identification and frequency domain fitting are compared and analyzed in Section 4, and finally, Section 5 concludes the paper.

2. Experimental study of EIS

The EIS provides an effective way to investigate the behavior of the energy storage devices in a non-destructive manner. The characteristics of the impedance spectroscopy can provide insights into electrochemical systems, which can be used to build mathematical models. By characterizing the impedance over wide frequency ranges, the EIS can distinguish the physical and electrochemical processes of the energy storage devices. In this section, the experiments of EIS testing for lithium-ion batteries and ultra-capacitors will be introduced.

2.1. Experimental setup

The type of the lithium-ion battery under test is LFP-1665130 with a rated capacity of 12 Ah, and the type of the ultra-capacitor is Maxwell BCAP3000-P270 (2.7 V/3.0 Wh). The anode and cathode materials of the tested lithium-ion battery are lithium iron phosphate and graphite, respectively. The ultra-capacitor is a kind of electronic double-layer capacitors (EDLCs).

The experimental setup for extracting the frequency response of the energy storage devices under test is depicted in Fig. 1. The experimental setup consists of an electrochemical workstation Solarton 1260/1287 for the EIS testing, a temperature chamber for thermostatic control from $-40(\pm1)\,^{\circ}$C to $150(\pm1)\,^{\circ}$C, and a host computer for data acquisition and analysis. The signal frequency range of the electrochemical workstation is from 0.01 Hz to 10 kHz, and the amplitude range of the current is from 20nA to 2A. In this work, the amplitude of the alternating current (AC) sinusoidal signals for the EIS test is set to 1.2A.

2.2. EIS at different temperature

The impedance of an energy storage device around some steady or quasi-steady state can be determined using the following procedures

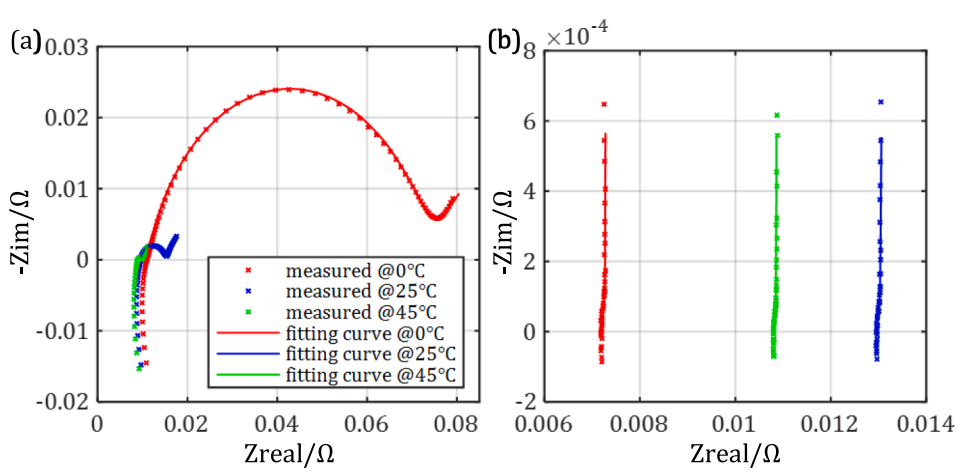

Fig. 2. EIS diagram for the impedance of the lithium-ion battery and ultra-capacitor at different temperatures: (a) lithium-ion battery (b) ultra-capacitor.

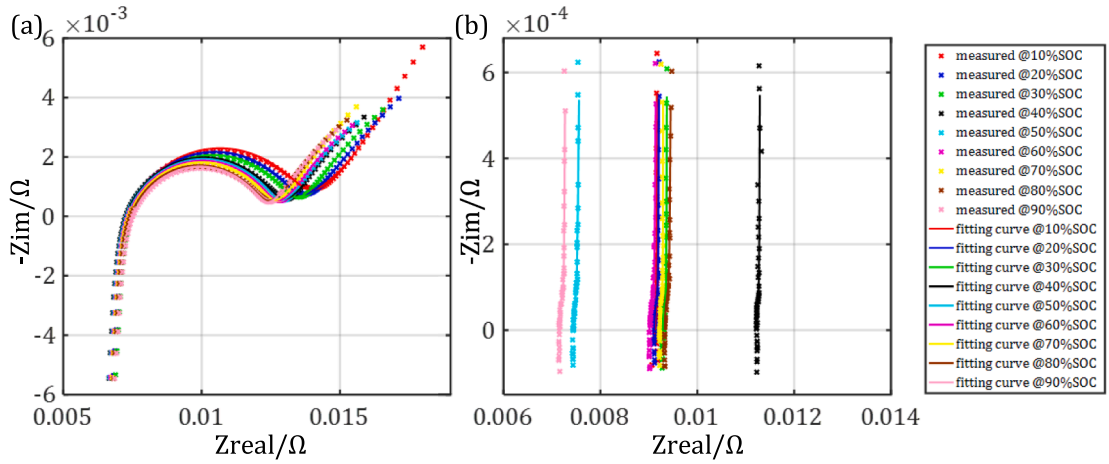

Fig. 3. EIS diagram for the impedance of the lithium-ion battery and ultra-capacitor at different SOC: (a) lithium-ion battery (b) ultra-capacitor.

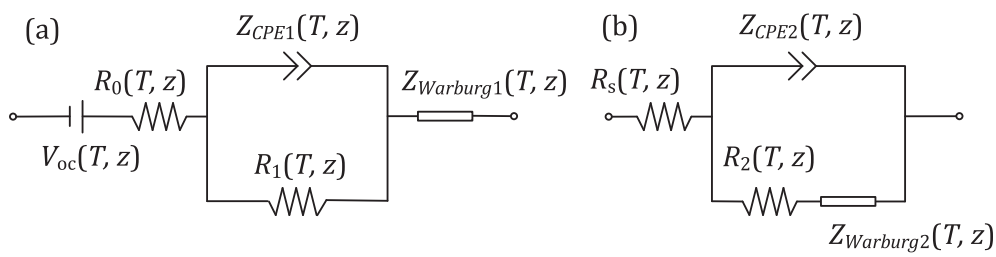

Fig. 4. Structure of the fractional-order models considering temperature and SOC: (a) Lithium-ion battery. (b) Ultra-capacitor.

[14]. First, apply a sequence of small AC sinusoidal signals to excite the system, and then measure the voltage response. Second, calculate the impedance by dividing the voltage by its corresponding current as follows:

$$Z = |V|e^{j(wt+\varphi_v)}/|I|e^{j(wt+\varphi_I)} = |V|e^{j(\varphi_v - \varphi_I)}/|I| \tag{1}$$

The Nyquist plots of the impedance spectrum at different temperatures are illustrated in Fig. 2(a) and (b). A general understanding is that the intersection of the Nyquist plot with the real axis is known as the ohmic internal resistance. The mid-frequency semi-ellipse is caused by double-layer effects and charge transfer reactions. The low-frequency straight lines of both the lithium-ion batteries and ultra-capacitors are related to the diffusion phenomenon. For the lithium-ion batteries, the experimental results of EIS indicate that the temperature has great impacts on both the mid-frequency and low-frequency dynamics. Specifically, the radius of the mid-frequency semi-ellipse has a discernable increase when the temperature drops. Moreover, the intersection of the Nyquist curve and the real axis shifts to the right as the temperature drops which means the ohmic internal resistance increases with the decrease of temperature. For the ultra-capacitors, the temperature also has great impacts on the Nyquist curve especially the intersection of the curve and the real axis. As a result, the temperature has significant effects on the internal parameters of both the batteries and ultra-capacitors which should be considered as an important factor for the modeling of batteries and ultra-capacitors.

2.3. EIS at different SOC

The Nyquist plots of the impedance spectrum at different state-of-charge (SOC) are illustrated in Fig. 3(a) and (b). From the experimental results, the remaining capacity has a certain effect on the impedance spectrum of both the lithium-ion battery and ultra-capacitor. At different SOC levels, the ohmic impedance of the lithium-ion battery changes very little, but the charge transfer impedance and diffusion impedance in the mid-frequency and low-frequency bands are significantly affected by the SOC. Compared with lithium-ion batteries, the effect of SOC on ultra-capacitor impedance is not obvious. For example, the impedance has a sudden increase at 40% SOC and drops when SOC equals to 50% and 90%. Even though the impedance of the ultra-capacitor is small, those impedance changes will directly affect the power output of the ultra-capacitor. Therefore, it is necessary to investigate management or control strategy to enable the ultra-capacitor to operate in an efficient SOC operating range.

3. Fractional-order models for lithium-ion battery and ultra-capacitor

The impedance spectroscopy not only reflects the kinetics of the energy storage device but also includes the mechanism of the electrode material, solid electrolyte, conductive polymer, and corrosion protection. In this section, the fractional-order models are developed based on the impedance spectroscopy of the lithium-ion batteries and ultra-capacitors.

3.1. Model description

The fractional-order model can be established by selecting the components which can represent the characteristics of the impedance

Table 1
Polynomial parameters of the OCV-SOC function.

Parameters	Value
k_0	$0.00003133T^2 - 0.001143T + 3.422$
k_1	$0.00001863\ T^2 - 0.0009858T + 0.05482$
k_2	$0.00003953\ T^2 + 0.002332T - 0.1792$
k_3	$0.00001323\ T^2 - 0.0009258T + 0.01488$
k_4	$0.0000679\ T^2 - 0.004375T + 0.1478$
k_5	$0.00001509\ T^2 - 0.0007311T - 0.006306$

spectroscopy in different frequency bands.

3.1.1. Fractional-order model for lithium-ion batteries

Based on the EIS results, the fractional-order model for lithium-ion batteries can be developed as shown in Fig. 4(a). The linear characteristics of the low-frequency band can be characterized by the Warburg-like element. The semi-elliptic characteristics of the mid-frequency band can be described by a CPE in parallel with a resistor. The intersection point of the Nyquist curve and real axis in the high-frequency band represents the equivalent internal ohmic resistance of the lithium-ion battery. From the experimental study of EIS in Section 2, the temperature and SOC have significant influences on the behavior of the battery. Therefore, the temperature and SOC are considered as variables of the model parameters in order to enhance the model accuracy and adaptability.

The Laplace transformation of the impedance of the CPE component is [14]:

$$Z_{CPE}(T,z) = \frac{1}{C(T,z)(j2\pi f)^{\alpha(T,z)}} \tag{2}$$

where T and z denote the temperature and SOC, C characterizes the capacitive reactance property of CPE components, f represents the frequency, j^2 equals to -1, α ($\alpha \in R^+$, $0 \le \alpha \le 1$) represents the fractional order of CPE. Note that when α equals to 1, the CPE characterizes the ideal capacitor, and when α equals to 0, the CPE denotes an ideal resistance.

Similarly, the Laplace transformation of the impedance of the Warburg-like component can be expressed as:

$$Z_{Warburg}(T,z) = \frac{1}{W(T,z)(j2\pi f)^{\beta(T,z)}} \tag{3}$$

where β ($\beta \in R^+$, $0 \le \alpha \le 1$) represents the fractional order of the Warburg-like component.

The transfer function of the battery model in the frequency domain can be expressed as follows:

$$\begin{aligned}\frac{Y_1(s)}{I_1(s)} &= \frac{V_{oc}(s) - V_T(s)}{I(s)} \\ &= R_0(T,z) + \frac{R_1(T,z)}{1 + C_1(T,z)R_1(T,z)s^{\alpha_1(T,z)}} + \frac{1}{W_1(T,z)s^{\beta_1(T,z)}}\end{aligned} \tag{4}$$

where $s = j2\pi f$, V_{oc} and V_T represent the battery open-circuit voltage (OCV) and terminal voltage, respectively.

Note that as an important part of the battery model, the OCV is also related to battery SOC and the operation temperature, which can be characterized by the following nonlinear function:

$$V_{oc}(T,z) = k_0(T) + k_1(T)z^2 + k_2(T)z + k_3(T)/z + k_4(T)log(z) + k_5(T)log(1-z) \tag{5}$$

where k_0 to k_5 are polynomial parameters of the nonlinear OCV-SOC function.

The identification results of the polynomial parameters using the least-squares method are shown in Table 1.

3.1.2. Fractional-order model for ultra-capacitors

Similarly, the fractional-order model for the ultra-capacitor is established as shown in Fig. 4(b) referring to the EIS of ultra-capacitors. The high-frequency band represents the equivalent series internal resistance. The characteristics of the low-frequency band and mid-frequency band can be described by a CPE in parallel with a resistor and a Warburg-like element. The transfer function of the ultra-capacitor's fractional-order model in the frequency domain can be expressed as:

$$\frac{Y_2(s)}{I_2(s)} = R_s(T,z) + \frac{1 + R_2(T,z)W_2(T,z)s^{\beta_2(T,z)}}{C_2(T,z)s^{\alpha_2(T,z)} + W_2(T,z)s^{\beta_2(T,z)} + R_2(T,z)C_2(T,z)W_2(T,z)s^{(\alpha_2(T,z)+\beta_2(T,z))}} \tag{6}$$

3.2. Parameter identification using global optimization

The EIS can obtain accurate model parameters. However, it requires specific signals to excite the system. In order to apply the fractional-order models in a real system and identify the model parameters on-line with the operation data, the parameter identification in time domain is necessary. Here the on-line parameter identification using global optimization is presented. Thus the parameters of the presented fractional-order models can be extracted from daily vehicle operating conditions.

By means of the fractional-order calculus, the transfer functions of the battery and ultra-capacitor models can be converted to fractional-order differential equations:

$$\begin{aligned}&(W_1(T,z)\mathfrak{D}^{\beta_1(T,z)} + C_1(T,z)R_1(T,z)W_1(T,z)\mathfrak{D}^{\alpha_1(T,z)+\beta_1(T,z)})Y_1(t) \\ &= (1 + C_1(T,z)R_1(T,z)\mathfrak{D}^{\alpha_1(T,z)} + (R_1(T,z)W_1(T,z) + R_0(T,z)W_1(T,z))\mathfrak{D}^{\beta_1(T,z)} \\ &\quad + R_0(T,z)C_1(T,z)R_1(T,z)W_1(T,z)\mathfrak{D}^{\alpha_1(T,z)+\beta_1(T,z)})I_1(t)\end{aligned} \tag{7}$$

$$\begin{aligned}&(C_2(T,z)\mathfrak{D}^{\alpha_2(T,z)} + W_2(T,z)\mathfrak{D}^{\beta_2(T,z)} + R_2(T,z)C_2(T,z)W_2(T,z)\mathfrak{D}^{(\alpha_2(T,z)+\beta_2(T,z))})Y_2(t) \\ &= (1 + (R_2(T,z)W_2(T,z) + R_s(T,z)W_2(T,z))\mathfrak{D}^{\beta_2(T,z)} + R_s(T,z)C_2(T,z)\mathfrak{D}^{\alpha_2(T,z)} \\ &\quad + R_s(T,z)R_2(T,z)C_2(T,z)W_2(T,z)\mathfrak{D}^{(\alpha_2(T,z)+\beta_2(T,z))})I_2(t)\end{aligned} \tag{8}$$

where $\mathfrak{D}^{\mu}$ denotes the fundamental operator of the fractional-order calculus and μ denotes the integral–differential order. Note that when $\mu > 0$, $\mathfrak{D}^{\mu}$ means fractional derivative; when $\mu < 0$, $\mathfrak{D}^{\mu}$ represents fractional integral; when $\mu = 0$, $\mathfrak{D}^{\mu}$ equals to 1. The mathematical expression is as follows [28]:

Table 2
EIS fitting results of battery model at different temperatures (SOC = 50%, #B1).

	$T = 0\ ^\circ C$	$T = 25\ ^\circ C$	$T = 45\ ^\circ C$
R_0(mΩ)	11.015	9.56	8.937
R_1(mΩ)	62.155	5.566	0.786
C_1(F/s$^{\alpha 1}$)	1.479	2.705	19.53
α_1	0.829	0.801	0.705
W_1(F/s$^{\beta 1}$)	400	1275	2100
β_1	0.552	0.615	0.572

Table 3
EIS fitting results of ultra-capacitor model at different temperatures (SOC = 50%, #C1).

	$T = 0\ ^\circ C$	$T = 25\ ^\circ C$	$T = 45\ ^\circ C$
R_s(mΩ)	7.191	12.96	10.796
R_2(mΩ)	0.167	0.173	0.141
C_2(F/s$^{\alpha 2}$)	892	1035	993
α_2	0.990	0.925	0.994
W_2(F/s$^{\beta 2}$)	1928	1791	1896
β_2	0.987	0.99	0.994

$$\mathfrak{D}^{\mu} = \begin{cases} d^{\mu}/dt^{\mu} & \mu > 0, \\ 1 & \mu = 0, \\ \int_{t_0}^{t} (d\tau)^{-\mu} & \mu < 0. \end{cases} \tag{9}$$

Based on the Grünwald-Letnikov definition [30], Eqs. (7) and (8) can be written as:

$$\begin{aligned} &\sum_{j=0}^{Nfom} (-1)^j \left[\frac{W_1(T,z)}{(\Delta t)^{\beta_1(T,z)}} \binom{\beta_1(T,z)}{j} + \frac{C_1(T,z)R_1(T,z)W_1(T,z)}{(\Delta t)^{\alpha_1(T,z)+\beta_1(T,z)}} \binom{\alpha_1(T,z)+\beta_1(T,z)}{j} \right] Y_1(t - j\Delta t) \\ &= I_1(t) + \sum_{j=0}^{Nfom} (-1)^j \left[\frac{C_1(T,z)R_1(T,z)}{(\Delta t)^{\alpha_1(T,z)}} \binom{\alpha_1(T,z)}{j} + \frac{R_1(T,z)W_1(T,z) + R_0(T,z)W_1(T,z)}{(\Delta t)^{\beta_1(T,z)}} \binom{\beta_1(T,z)}{j} \right. \\ &\qquad \left. + \frac{R_0(T,z)C_1(T,z)R_1(T,z)W_1(T,z)}{(\Delta t)^{\alpha_1(T,z)+\beta_1(T,z)}} \binom{\alpha_1(T,z)+\beta_1(T,z)}{j} \right] I_1(t - j\Delta t) \end{aligned} \tag{13}$$

$$\begin{aligned} &\sum_{j=0}^{Nfom} (-1)^j \left[\frac{C_2(T,z)}{(\Delta t)^{\alpha_2(T,z)}} \binom{\alpha_2(T,z)}{j} + \frac{W_2(T,z)}{(\Delta t)^{\beta_2(T,z)}} \binom{\beta_2(T,z)}{j} \right. \\ &\qquad \left. + \frac{R_2(T,z)C_2(T,z)W_2(T,z)}{(\Delta t)^{\alpha_2(T,z)+\beta_2(T,z)}} \binom{\alpha_2(T,z)+\beta_2(T,z)}{j} \right] Y_2(t - j\Delta t) \\ &= I_2(t) + \sum_{j=0}^{Nfom} (-1)^j \left[\frac{(R_2(T,z)W_2(T,z) + R_s(T,z)W_2(T,z)}{(\Delta t)^{\beta_2(T,z)}} \binom{\beta_2(T,z)}{j} + \frac{R_s(T,z)C_2(T,z)}{(\Delta t)^{\alpha_2(T,z)}} \binom{\alpha_2(T,z)}{j} \right. \\ &\qquad \left. + \frac{R_s(T,z)R_2(T,z)C_2(T,z)W_2(T,z)}{(\Delta t)^{\alpha_2(T,z)+\beta_2(T,z)}} \binom{\alpha_2(T,z)+\beta_2(T,z)}{j} \right] I_2(t - j\Delta t) \end{aligned} \tag{14}$$

The commonly used definitions of the fractional-order calculus include the Riemann-Liouville definition, the Grünwald-Letnikov definition, the Caputo definition, and the Cauchy definition [29]. The specific definition of Grünwald-Letnikov which defines how to directly discretize continuous fractional differential equations is shown as follows [30]:

$$\mathfrak{D}^{\mu} f(t) = \lim_{h \to 0} \frac{1}{\Delta t^{\mu}} \sum_{j=0}^{[(t-t_0)/\Delta t]} (-1)^j \binom{\mu}{j} f(t - j\Delta t) \tag{10}$$

where $[(t\text{-}t_0)/\triangle t]$ represents the integer part of $(t\text{-}t_0)/\triangle t$, and $\binom{\mu}{j}$ can be expressed by:

$$\binom{\mu}{j} = \frac{\Gamma(\mu+1)}{\Gamma(j+1)\Gamma(\mu-j+1)} \tag{11}$$

$$\Gamma(\mu) = \int_0^{\infty} t^{\mu-1} e^{-t} \mathrm{d}t \tag{12}$$

Then the time domain identification can be applied to Eqs. (13) and (14). In this work, the global optimization using particle swarm optimization (PSO) algorithm is applied [31]. The system inputs I_1 and I_2 are the currents of the battery and ultra-capacitor. The output Y_1 denotes the difference between the battery open-circuit voltage and terminal voltage. The output Y_2 denotes the terminal voltage of the ultra-capacitor. The parameter vectors of the battery and ultra-capacitor to be identified are as follows:

$$\theta_1(T,z) = [R_0(T,z), R_1(T,z), C_1(T,z), \alpha_1(T,z), W_1(T,z), \beta_1(T,z)]^T \tag{15}$$

$$\theta_2(T,z) = [R_s(T,z), R_2(T,z), C_2(T,z), \alpha_2(T,z), W_2(T,z), \beta_2(T,z)]^T \tag{16}$$

The problem aims to find the optimal parameters to minimize the error between the model and the measurements:

$$\hat{\theta}_i = \underset{\theta_i}{argmin}\left(\sqrt{\sum\nolimits_{k=Nfom+1}^{N} (Y_i(k) - \hat{Y}_i(k,\theta_i))^2/(N - Nfom)}\right), (i = 1, 2) \tag{17}$$

Table 4
EIS fitting results of battery model at different SOC ($T = 25$ °C, #B2).

	10%	20%	30%	40%	50%	60%	70%	80%	90%
R_0(mΩ)	7.37	7.38	7.45	7.255	7.3	7.33	7.375	7.43	7.52
R_1(mΩ)	6.243	5.9454	5.7473	5.5246	5.388	5.267	5.133	4.815	4.718
C_1(F/s$^{\alpha 1}$)	3.073	2.867	2.925	2.923	2.929	2.967	3.061	3.078	3.111
α_1	0.782	0.784	0.779	0.779	0.779	0.777	0.774	0.775	0.774
W_1(F/s$^{\beta 1}$)	608.1	761.8	924.2	1032	1107	1142	1042	1051	1190
β_1	0.510	0.494	0.524	0.531	0.537	0.540	0.565	0.538	0.542

Table 5
EIS fitting results of ultra-capacitor model at different SOC ($T = 25$ °C, #C2).

	10%	20%	30%	40%	50%	60%	70%	80%	90%
R_s(mΩ)	9.13	9.114	9.29	11.22	7.435	9.015	9.217	9.327	7.149
R_2(mΩ)	0.093	0.155	0.144	0.0142	0. 216	0. 212	0. 147	0.199	0. 175
C_2(F/s$^{\alpha 2}$)	1150	892.5	908.6	1078	1242	1038	886.4	1030	830.9
α_2	0.996	0.956	0.954	0.99	0.854	0.863	0.98	0.906	0.946
W_2(F/s$^{\beta 2}$)	1744	2037	1999	1826	1601	1854	2100	1966	2290
β_2	0.995	0.986	0.995	0.977	0.996	0.992	0.999	0.976	0.975

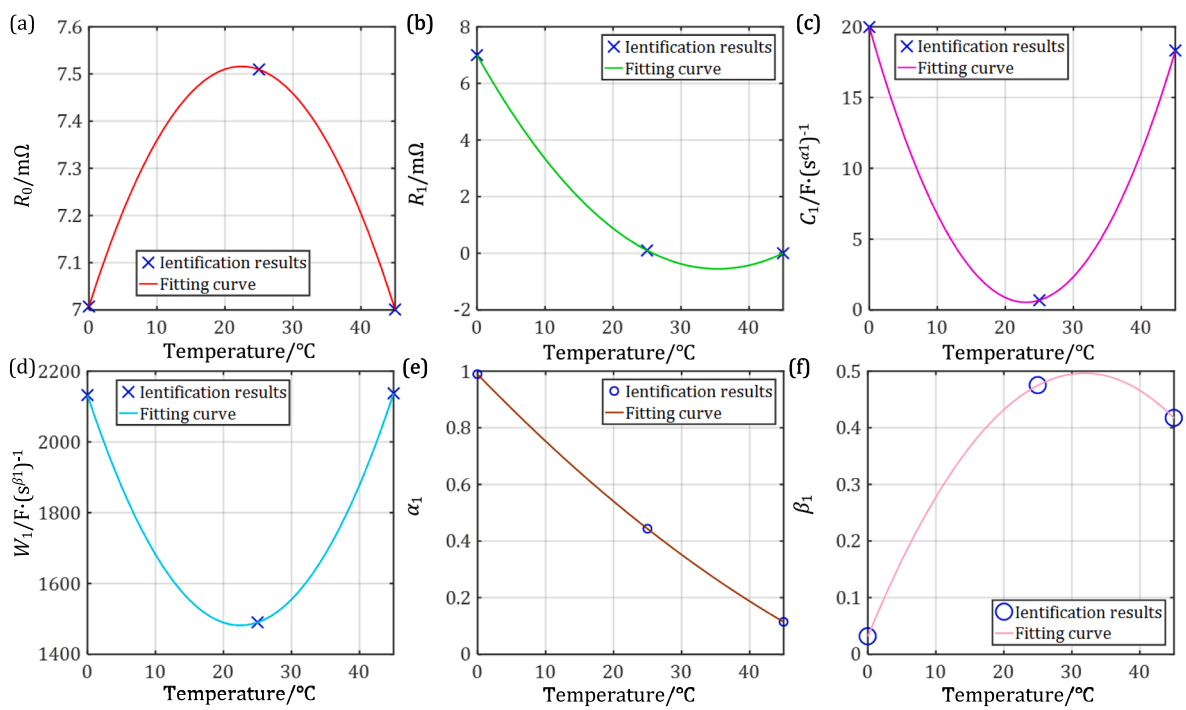

Fig. 5. Model parameter identification results of the lithium-ion batteries.

where N denotes the length of the training dataset.

4. Results and discussion

In order to verify the proposed fractional-order models and the parameter identification algorithm, results and discussion will be presented in this section.

4.1. Off-line identification based on EIS

The EIS experiments introduced in Section 2 are used to conduct experiments on the lithium iron phosphate batteries and ultra-capacitors, in order to establish constraints for the PSO algorithm. The Matlab toolbox is used for curve fitting of the impedance frequency response data. The model parameters of the lithium-ion battery (Cell #B1) and ultra-capacitor (Cell #C1) at different temperatures based on off-line EIS data fitting are shown in Tables 2 and 3. The model parameters of the lithium-ion battery (Cell #B2) and ultra-capacitor (Cell

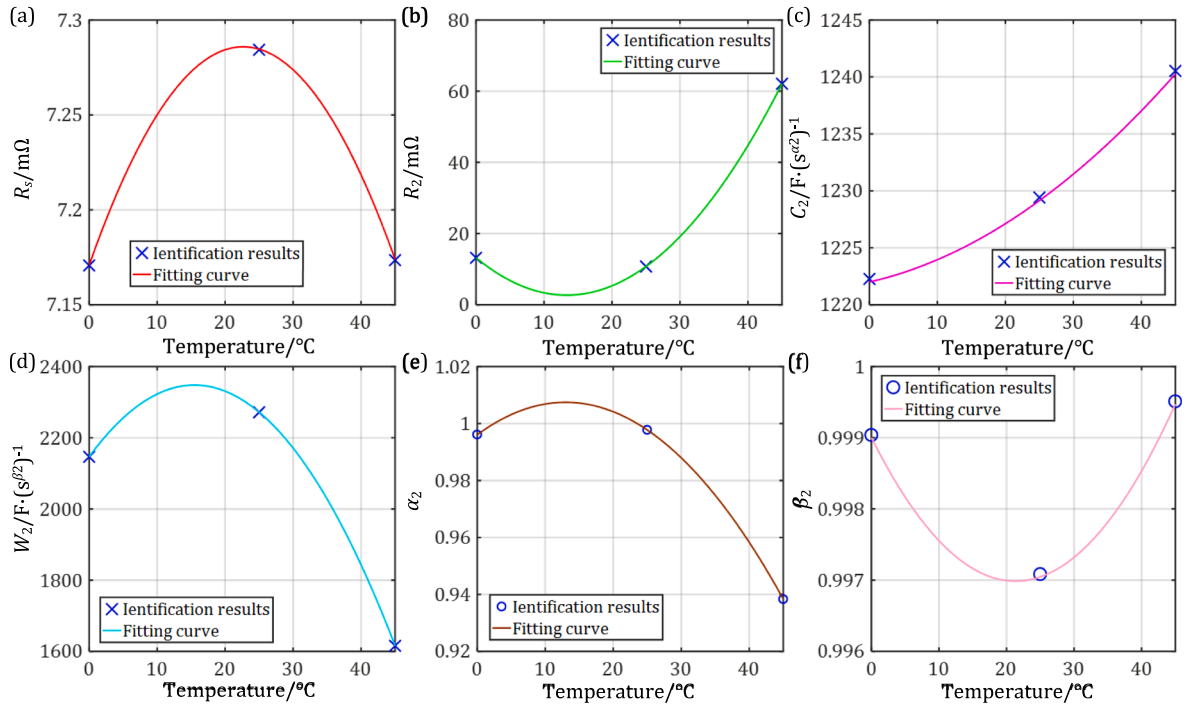

Fig. 6. Model parameter identification results of the ultra-capacitors.

#C2) at different SOC based on the EIS data fitting are shown in Tables 4 and 5. These results can be used as benchmarks for parameter identification, and constraints for global optimization.

4.2. Parameter identification based on PSO

To avoid over-fitting in global optimization processes, the configuration of the constraint boundaries are significant. The constraints can be roughly determined by the off-line identification results of EIS tests. Based on the EIS testing results of the lithium-ion batteries and ultra-capacitors, the constraints of the battery model are as follows: $R_0 \in [7\ \text{m}\Omega, 12\ \text{m}\Omega]$, $R_1 \in [0\ \text{m}\Omega, 70\ \text{m}\Omega]$, $C_1 \in [0\ \text{F/s}^{\alpha 1}, 20\ \text{F/s}^{\alpha 1}]$, $\alpha_1 \in [0, 1]$, $W_1 \in [400\ \text{F/s}^{\beta 1}, 2200\ \text{F/s}^{\beta 1}]$, $\beta_1 \in [0, 1]$, and the constraints of the ultra-capacitor model are as follows: $R_s \in [7\ \text{m}\Omega, 13\ \text{m}\Omega]$, $R_2 \in [10\ \text{m}\Omega, 200\ \text{m}\Omega]$, $C_2 \in [800\ \text{F/s}^{\alpha 2}, 1250\ \text{F/s}^{\alpha 2}]$, $\alpha_2 \in [0, 1]$, $W_2 \in [1600\ \text{F/s}^{\beta 2}, 2300\ \text{F/s}^{\beta 2}]$, $\beta_2 \in [0, 1]$.

In this work, the training datasets used for parameter identification of the presented fractional-order models are from the federal urban driving schedule and the dynamic stress test at different temperatures. The testing datasets for model validation are from the urban dynamometer driving schedule at 0 °C, 25 °C, and 45 °C in full SOC ranges. The parameter identification results of the fractional-order models for the lithium-ion batteries and ultra-capacitors at different temperatures are plotted in Figs. 5 and 6.

The prediction results of the fractional-order models for the lithium-ion battery and ultra-capacitor at different temperatures are plotted in Figs. 7 and 8. The numerical results of the prediction performance including the mean absolute error (MAE), mean relative error (MRE), and root-mean-square error (RMSE) are shown in Tables 6 and 7, respectively. The MAEs of the fractional-order model for the lithium-ion batteries at 0 °C, 25 °C, and 45 °C are 93.2 mV, 12.9 mV, and 22.2 mV, respectively. The RMSEs of the presented battery model at 0 °C, 25 °C, and 45 °C are 124.9 mV, 18.0 mV, and 45.3 mV. Correspondingly, the MREs of the fractional-order model for the lithium-ion batteries at 0 °C, 25 °C, and 45 °C are 3.09%, 0.41%, and 0.72%. The results indicate that the model prediction of the fractional-order model at 25 °C, and 45 °C in full SOC ranges can well approximate the terminal voltage of the lithium-ion batteries. At low-temperature conditions, the prediction results of the battery model show some deviation in low SOC ranges. This is mainly because the internal resistances of the lithium-ion batteries at low temperatures are higher than that at high temperatures. The battery quickly reaches the cut-off voltage when a large current occurs.

For the fractional-order model of the ultra-capacitors, the MAEs at 0 °C, 25 °C, and 45 °C are 26.4 mV, 37.3 mV, and 28.2 mV, respectively. The RMSEs of the fractional-order model for the ultra-capacitors at 0 °C, 25 °C, and 45 °C are 32.8 mV, 45.1 mV, and 35.4 mV, respectively. The MREs of the model at 0 °C, 25 °C, and 45 °C conditions are 1.45%, 2.1%, and 1.52%. The results indicate that the presented fractional-order models and PSO-based model parameter identification algorithm can well approximate the voltage of the ultra-capacitors with MAEs and RMSEs less than 40 mV and 50 mV, respectively. Moreover, the mean relative errors are less than 3%. As a result, the proposed fractional-order model can describe the voltage characteristics of the lithium-ion batteries in full SOC ranges at different temperatures.

5. Conclusions

Accurate modeling of lithium-ion batteries and ultra-capacitors is the foundation of power and energy management in hybrid energy storage systems. Compared with the conventional electrochemical models and equivalent circuit models, the fractional-order models have unique advantages in computational complexity and approximating accuracy. In this work, the experimental study of fractional-order models for both the lithium-ion batteries and ultra-capacitors is presented. The original contributions of this work can be summarized as follows: (1) The effects of temperature and state-of-charge on the lithium-ion battery and ultra-capacitor impedance spectra are systematically analyzed based on the experiments of electrochemical impedance spectroscopy. (2) The

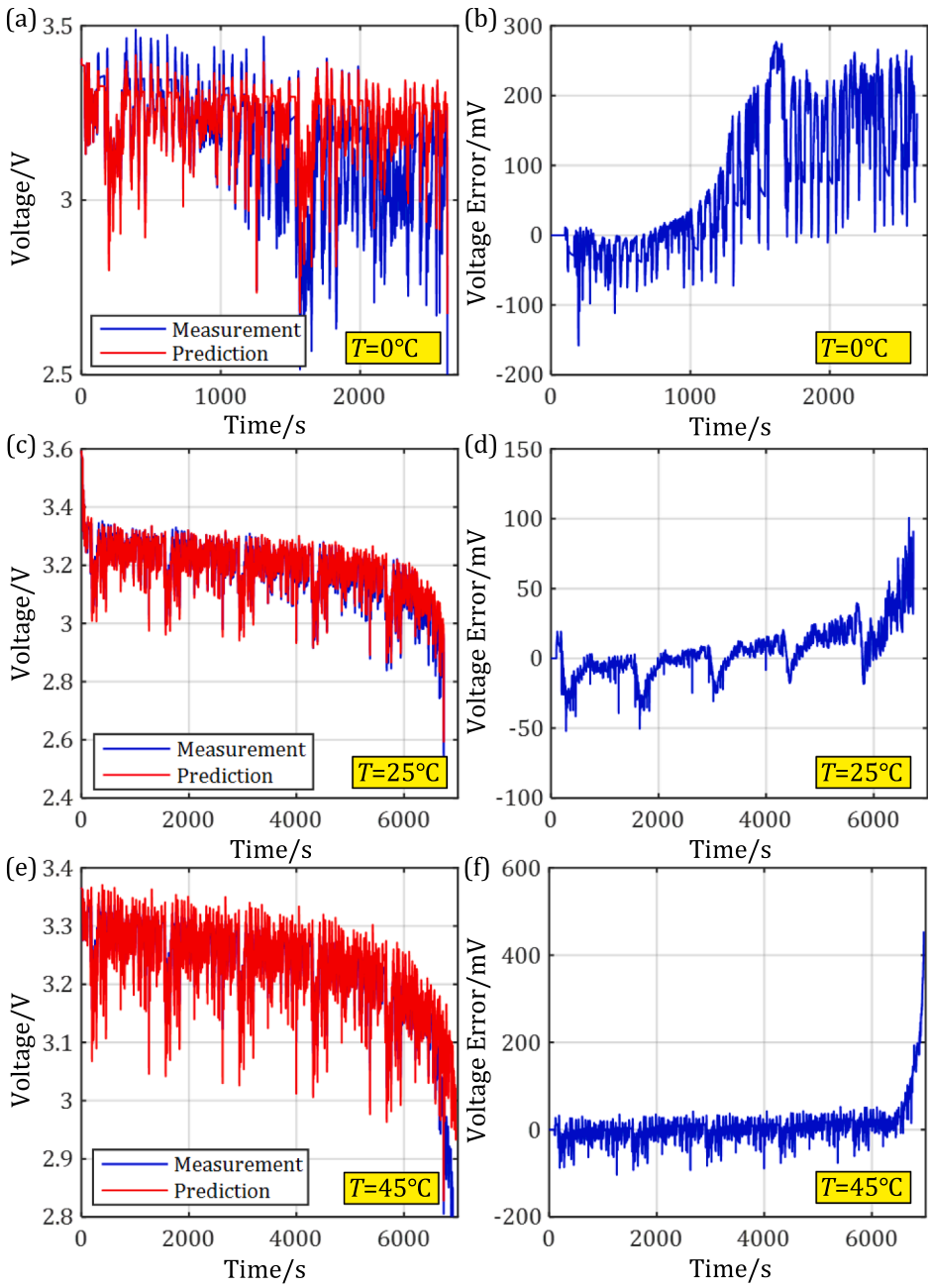

Fig. 7. Model prediction results of the fractional-order model for lithium-ion batteries at different temperatures: (a) Terminal voltage at 0 °C. (b) Voltage error at 0 °C. (c) Terminal voltage at 25 °C. (d) Voltage error at 25 °C. (e) Terminal voltage at 45 °C. (f) Voltage error at 45 °C.

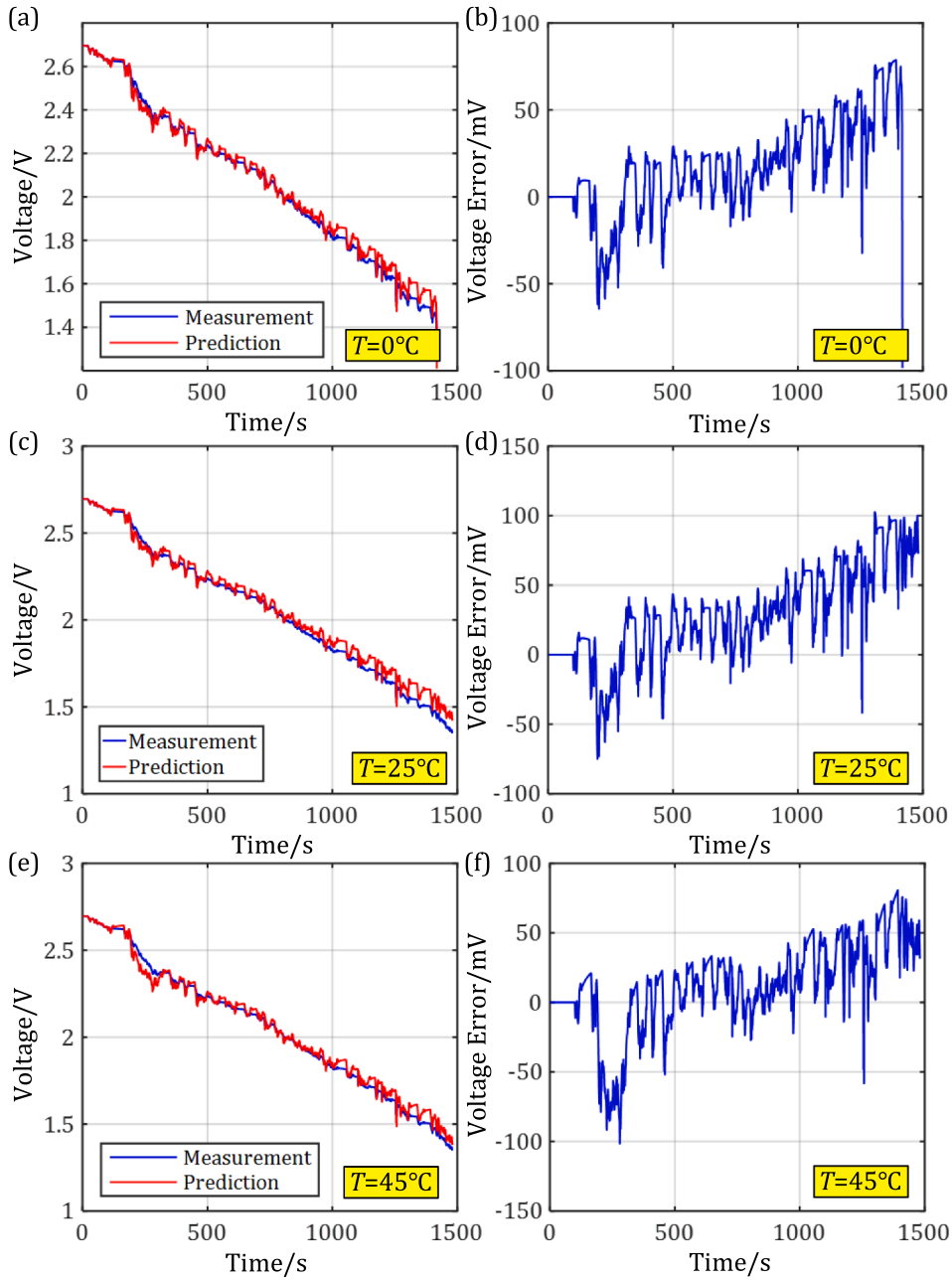

Fig. 8. Model prediction results of the fractional-order model for ultra-capacitors at different temperatures: (a) Terminal voltage at 0 °C. (b) Voltage error at 0 °C. (c) Terminal voltage at 25 °C. (d) Voltage error at 25 °C. (e) Terminal voltage at 45 °C. (f) Voltage error at 45 °C.

Table 6
Performance of the battery model at different temperatures.

	$T = 0$ °C	$T = 25$ °C	$T = 45$ °C
MAE[a]	93.2 mV	12.9 mV	22.2 mV
MRE[b]	3.09%	0.41%	0.72%
RMSE[c]	124.9 mV	18.0 mV	45.3 mV

[a] MAE: mean absolute error; [b]MRE: mean relative error; [c]RMSE: root-mean-square error.

Table 7
Performance of the ultra-capacitor model at different temperatures.

	$T = 0$ °C	$T = 25$ °C	$T = 45$ °C
MAE	26.4 mV	37.3 mV	28.2 mV
MRE	1.45%	2.10%	1.52%
RMSE	32.8 mV	45.1 mV	35.4 mV

fractional-order models for the lithium-ion batteries and ultra-capacitors are developed taking into consideration of both temperature and state-of-charge. (3) The particle swarm optimization based global optimization is proposed for on-line parameter identification with strict constraints. Thus the parameters of the fractional-order models can be extracted from daily vehicle operating conditions. (4) The accuracy of the presented model is proved by dynamic cycle conditions at different temperatures. The results indicate that the proposed fractional-order models can well simulate the voltage with high accuracy.

Though the fractional-order model proposed in this paper has high fidelity, the population size and boundary conditions of particle swarm optimization algorithm have great influence on parameter identification. Therefore, our future work will focus on further improvement of

the fidelity of the fractional-order models and the performance of the particle swarm optimization algorithm. In addition, the presented models will be applied in state estimation and energy management of hybrid energy storage systems to further improve the performance of the control algorithms.

CRediT authorship contribution statement

Yujie Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft. **Mince Li:** Conceptualization, Writing - review & editing. **Zonghai Chen:** Resources, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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